

Distributed Optimal Fault Estimation and Fault-Tolerant Control for Interconnected Systems: A Stackelberg Differential Graphical Game Approach

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Abstract—This article addresses the distributed optimal fault estimation (FE) and fault-tolerant control (FTC) issues under the bidirectional interaction between a possibly faulty interconnected system and its associated diagnostic interconnected observer. The proposed methodology is based on a framework, named *Stackelberg differential graphical game*, in which the interconnected observer and its corresponding interconnected system are formulated as two graphical games, and they, respectively, play the role of the follower and leader to make *sequential* decisions of the FE and the FTC. Such a framework allows us to fully reveal the property of the bidirectional interaction, based on which a distributed optimal FE and FTC scheme is developed. Moreover, a new concept named *interactive Stackelberg equilibrium* is proposed, where, to reach this equilibrium, an auxiliary controller is suitably designed for each subobserver. In this way, this auxiliary controller can cooperate with other auxiliary controllers of subobservers, and noncooperate with controllers of subsystems, which helps to achieve the distributed optimal FE and FTC goals.

Index Terms—Fault estimation (FE), fault-tolerant control (FTC), graphical game, interconnected systems, Stackelberg differential game.

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I. INTRODUCTION

Interconnected systems are composed of several subsystems that are coupled with each other [1]. Such systems have broad applications, such as smart grids [2] and aircraft formation [3]. In practice, interconnected systems are usually exposed to complex environments, which may result in different types of faults.

Active fault-tolerant control (FTC), which is also referred to as fault accommodation, is based on the reconfiguration of the controller in response to the detection of occurred faults [4]–[7]. To achieve the design of fault estimation (FE) and FTC independently in active FTC, the classic separation principle is usually pursued [8]. However, due to the bidirectional interaction between the system and the observer in the case of mismatch, interconnection, and estimation errors, the separation principle often breaks down [9]. Therefore, revealing the relation between the FE and the FTC, and designing them more effectively are still challenging problems.

In the case of designing the FE and FTC for interconnected systems, two main approaches may be followed: 1) Make the separation principle work either locally or globally. The main idea here is to let the separation principle take effect locally in the individual subsystem while the stability of the overall interconnected system is maintained by an appropriately connected topology [10], [11]. To apply the separation principle globally, an interconnected separation principle is proposed in [12]. Note that the aforementioned methods have some specific requirements on the connected topology. 2) Integrate the design of the FE and FTC no matter the separation principle is satisfied or not [9], [13]. The main idea of an integrated design is to combine the FE and FTC modules such that the parameters of the FE and FTC can be calculated simultaneously in a single linear matrix inequality. However, as a result of this integrated design, the dynamical bidirectional interaction property between the interconnected system and its associated observer cannot be fully revealed. Moreover, under such a bidirectional interaction, achieving the optimal FE and FTC performances for the overall interconnected system is a challenge, especially in a distributed communication framework [14].

The graphical game describes the players connecting with each other through a communication graph, which also forms the interest network of players such that one player's interest relies on the strategies of its own and its neighbors [15]. On the other hand, the Stackelberg differential game consists of two players: the leader and the follower [16], [17]. The key characteristic of this game is a sequential decision framework, in which the leader has a priority in announcing a strategy before the follower. After the follower receives the current strategy of the leader, it responds and tries to optimize its own performance. Meanwhile, the leader is aware of this situation, so that it tries to select its best response to the follower's best response. Due to this sequential decision feature, the Stackelberg differential game approach has been used in several applications, such as the smart grid [18] and the design of defense strategies against attacks [20]. Inspired by such a two-player sequential decision idea, in this article, we consider whether we can regard the controlled system and diagnostic observer as two players in a framework that allows the application of the Stackelberg differential

game and graphical game to solve the optimal FE and FTC problem of the interconnected systems.

This article aims at giving a positive answer to the previous question. The distributed optimal FE and FTC problems for interconnected systems are addressed in a framework named Stackelberg differential graphical game, which allows us to investigate the bidirectional interaction between the interconnected system and its associated observer from the decision perspective. This viewpoint guides us to develop a distributed optimal FE and FTC scheme. The main contributions of this article are summarized as follows.

- 1) A novel analysis of distributed FE and FTC behaviors is presented by making sequential decisions instead of following an integrated design [9], [13]. This helps to clearly show the bidirectional interaction property between the faulty interconnected system and its associated diagnostic interconnected observer, which is the basis of the sequential FE and FTC scheme. Moreover, it relieves the constraints on the topology compared with the work in [10]–[12].
- 2) A distributed optimal FE and FTC scheme is developed for the interconnected system based on the Stackelberg differential graphical game, in which the diagnostic interconnected observer acts as a follower to achieve the optimal FE, whereas the interconnected system acts as a leader to achieve the optimal FTC. To the best of our knowledge, this is the first attempt to solve the optimal active FTC problem in a Stackelberg game framework. It is proved that the faulty interconnected system and its associated diagnostic interconnected observer can reach the interactive Stackelberg equilibrium under the designed optimal FTC strategy.
- 3) An auxiliary controller is introduced and acts as the decision maker for each subobserver such that it can cooperate with other subobservers' auxiliary controllers to achieve the Nash equilibrium in the follower-graphical game while noncooperating with the subsystems' controllers to achieve the interactive Stackelberg equilibrium. With the help of this design, both the optimal FE and FTC performances are achieved.

The remainder of this article is organized as follows: Section II presents the problem formulation. Section III describes the problem of distributed optimal FE and FTC into a Stackelberg differential graphical game. The designs for the leader and follower are presented in Sections IV and V, respectively. A numerical example is presented in Section VI, followed by conclusions in Section VII.

Notation: We denote by x^T (or A^T) the transposition of a vector x (or a matrix A), $0_{n \times m} \in R^{n \times m}$ is a matrix with all elements being zero, $I_n \in R^n$ is a column vector with all elements being 1, $I_{n \times n} \in R^{n \times n}$ is an identity matrix, and $M \triangleq \text{diag}(M_1, M_2)$ is a block matrix with diagonal matrices being M_1 and M_2 .

II. PROBLEM FORMULATION

A. Subsystem's Dynamics

Consider a linear interconnected system composed of m subsystems whose connected behavior is represented as a directed graph $\mathcal{G} \triangleq \{\mathcal{M}, \mathcal{C}\}$ with $\mathcal{M} \triangleq \{1, 2, \dots, m\}$ representing the set of nodes and $\mathcal{C}$ denoting the set of arcs. An arc from node j to node i is represented as (j, i) , which implies that subsystem j is a neighbor of subsystem i . The dynamics of each subsystem i are described by

$$\begin{aligned} \dot{z}_i(t) &= \bar{A}_i z_i(t) + \bar{B}_i u_i(t) + \sum_{j \in N_i} \bar{B}_{ij} u_j(t) + \bar{E}_i f_i(t) \\ &\quad + \sum_{j \in N_i} \bar{E}_{ij} f_j(t) + \sum_{j \in N_i} \bar{D}_{ij} z_j(t) \\ y_i(t) &= \bar{C}_i z_i(t) \end{aligned} \quad (1)$$

where $z_i \in R^{n_i}$, $u_i \in R^{q_i}$, and $y_i \in R^{p_i}$ are the state, the control input, and the output of subsystem i , respectively. The neighboring set of node i is denoted as $N_i \triangleq \{j : (j, i) \in \mathcal{C}\}$. The term $\bar{D}_{ij} z_j$ is the

interconnection, and $\bar{B}_{ij} u_j$ represents the coupled input with respect to the neighboring subsystem j , which widely exists in practice, such as in the case of multiarea power systems [28].

The fault signal $f_i \in R^{p_{f_i}}$ is generated from the output of a dynamical exosystem given by

$$\begin{aligned} \dot{x}_{f_i}(t) &= \bar{A}_{f_i} x_{f_i}(t), \quad t \geq t_f \\ f_i(t) &= \bar{C}_{f_i} x_{f_i}(t) \end{aligned} \quad (2)$$

where $x_{f_i} \in R^{n_{f_i}}$ is the fault-state with the initial fault value $x_{f_i}(0) = x_{f_{i0}}$ being an unknown value; t_f is the unknown time that the fault occurs. The term $\bar{E}_i$ is the fault distribution matrix, and $\bar{C}_{f_i}$ is the output matrix, which has full row rank. Since a fault that has occurred in one subsystem may spread throughout the overall interconnected system via the interconnections, the fault propagation $\bar{E}_{ij} f_j$ is included in (1) to represent the fault effect from neighboring subsystems. The terms $\bar{A}_i, \bar{A}_{f_i}, \bar{B}_i, \bar{B}_{ij}, \bar{C}_i, \bar{C}_{f_i}, \bar{D}_{ij}, \bar{E}_i$, and $\bar{E}_{ij}$ are known and bounded matrices with appropriate dimensions.

Remark 1: The fault model in (2) can be regarded as an alternative parameterization for a class of faults [21], [22]. For example, suppose $\bar{A}_{f_i} = 0$; then, the solution of the model (2) is $f_i(t) = \bar{C}_{f_i} e^{\bar{A}_{f_i} t} x_{f_i}(0) = \bar{C}_{f_i} x_{f_i}(0)$, which can parameterize constant faults such as stuck faults of the aircraft control surface [21]. Moreover, the model (2) can also be used for approximating more general classes of faults by setting $\bar{A}_{f_i} = \text{diag}\{\Lambda_1, \Lambda_2, \dots, \Lambda_{(1/2)n_{f_i}}\}$

with $\Lambda_s \triangleq \begin{bmatrix} 0 & w_s \\ -w_s & 0 \end{bmatrix}$, $s \in \bar{\chi} \triangleq \{1, 2, \dots, n_{f_i}\}$, $x_{f_i}(0) = [\theta_{11}, \theta_{21}, \theta_{12}, \theta_{22}, \dots, \theta_{1(1/2)n_{f_i}}, \theta_{2(1/2)n_{f_i}}]^T$, and $\bar{C}_{f_i} = [1, 1, \dots, 1]_{1 \times n_{f_i}}$; then, the faults can be represented by $f_i(t) = \sum_{s=1}^{1/2 n_{f_i}} ((\theta_{1s} + \theta_{2s}) \cos w_s t + (\theta_{2s} - \theta_{1s}) \sin w_s t)$, in which $\sin w_s t$ and $\cos w_s t$ provide the basis functions with the frequencies w_s for approximation. Thus, the model (2) can characterize faults with known frequencies but with unknown magnitudes, which are sometimes referred to as parametric faults [23]. The richer the frequencies, the more accurate the parametric approximation. Moreover, the fault model (2) can represent not only actuator faults (shown as the previous examples), but also sensor faults [24]. $\square$

Remark 2: In the case of unknown fault distribution matrix $\bar{E}_i$, fault isolation techniques can be used for determining the channels that faults have occurred [25]. Alternatively, a model-free Stackelberg differential game may also provide a useful tool to address the limitations resulting from the unknown $\bar{E}_i$. According to this method, an algorithm based on adaptive dynamic programming (ADP) is developed to obtain the gains of both the follower and the leader under certain conditions [26]. $\square$

The following assumption will be required throughout this article.

Assumption 1.

- i) Each pair $(\bar{A}_i, \bar{B}_i)$ is controllable.
- ii) The pairs $(\bar{A}_i, \bar{C}_i)$ and $(\bar{A}_{f_i}, \bar{C}_{f_i})$ are observable.
- iii) For any eigenvalue λ_{f_i} of the matrix $\bar{A}_{f_i}$, the matrix $\begin{bmatrix} \lambda_{f_i} I - \bar{A}_i & -\bar{E}_i \\ \bar{C}_i & 0_{p_i \times n_{f_i}} \end{bmatrix}$ has full column rank. $\square$

Remark 3: Condition (i) in Assumption 1 guarantees the controllability, and Conditions (ii) and (iii) guarantee the observability of the augmented systems (3). Assumption 1 is a standard one in the state-augmentation-based FTC field [9], [24]. $\square$

B. Two Kinds of Interactions

To estimate the state and fault simultaneously, the model of subsystem i is augmented by setting $x_i \triangleq [z_i^T, x_{f_i}^T]^T$, so that

$$S_i : \begin{cases} \dot{x}_i = A_i x_i + B_i u_i + \sum_{j \in N_i} B_{ij} u_j + \sum_{j \in N_i} D_{ij} x_j \\ y_i = C_i x_i \end{cases} \quad (3)$$

where

$$A_i \triangleq \begin{bmatrix} \bar{A}_i & \bar{E}_i \bar{C}_{fi} \\ 0_{n_{fi} \times n_i} & \bar{A}_{fi} \end{bmatrix}, B_i \triangleq \begin{bmatrix} \bar{B}_i \\ 0_{n_{fi} \times q_i} \end{bmatrix}, B_{ij} \triangleq \begin{bmatrix} \bar{B}_{ij} \\ 0_{n_{fi} \times q_j} \end{bmatrix}, D_{ij} \triangleq \begin{bmatrix} \bar{D}_{ij} & \bar{E}_{ij} \bar{C}_{fi} \\ 0_{n_{fi} \times n_j} & 0_{n_{fi} \times n_{fj}} \end{bmatrix}, \text{ and } C_i \triangleq [\bar{C}_i \ 0_{p_i \times n_{fi}}].$$

Under Assumption 1, a general distributed interconnected observer is constructed for the interconnected system (3), in which the subobserver i , $\forall i \in \mathcal{M}$, is described as

$$\tilde{O}_i : \begin{cases} \dot{\hat{x}}_i = A_i \hat{x}_i + B_i u_i + \sum_{j \in N_i} B_{ij} u_j + \sum_{j \in N_i} D_{ij} \hat{x}_j + \bar{L}_i (y_i - \hat{y}_i) \\ \hat{y}_i = C_i \hat{x}_i \end{cases} \quad (4)$$

where $\hat{x}_i$ and $\hat{y}_i$ are the estimates of x_i and y_i , respectively, and $\bar{L}_i$ is the feedback matrix to be designed.

The general cost functions for subsystem i and subobserver i are given, respectively, by

$$a) \bar{J}_i \triangleq \int_0^\infty \Omega_i(x_i, u_i) dt, \quad b) \tilde{J}_i \triangleq \int_0^\infty \chi_i(e_{x_i}) dt \quad (5)$$

where $e_{x_i} \triangleq x_i - \hat{x}_i$ is the estimation error, $\Omega_i(\cdot)$ and $\chi_i(\cdot)$ are continuous functions with respect to their variables.

The time derivative of e_{x_i} is obtained as

$$\dot{e}_{x_i} = (A_i - \bar{L}_i C_i) e_{x_i} + \sum_{j \in N_i} D_{ij} x_j - \sum_{j \in N_i} D_{ij} \hat{x}_j \quad (6)$$

which reflects the optimal FE performance of subobserver i with the optimal matrix $\bar{L}_i$ minimizing the cost function (5-b).

Based on the optimal FE, a general distributed optimal fault-tolerant controller takes the form of

$$u_i^* = -K_i \hat{x}_i - \sum_{j \in N_i} \bar{K}_{ij} \hat{x}_j \quad (7)$$

where K_i and $\bar{K}_{ij}$ are feedback matrices designed to minimize the cost function (5-a). Applying the optimal fault-tolerant controller u_i^* in (7) into the subsystem i in (3), the closed-loop dynamics of subsystem i satisfy

$$\begin{aligned} \dot{x}_i = & (A_i - B_i K_i) x_i + B_i K_i e_i + \sum_{j \in N_i} D_{ij} x_j \\ & - B_i \sum_{j \in N_i} \bar{K}_{ij} \hat{x}_j + \sum_{j \in N_i} B_{ij} u_j \end{aligned} \quad (8)$$

which reflects the optimal FTC performance of subsystem i .

It can be observed from (3), (4), (6), and (8) that there exist two kinds of interactions, which are as follows.

- 1) *Internal interaction*, which exists *inside* the interconnected system/observer. From (3) and (4), it can be seen that subsystem i is influenced by its neighboring subsystem j through x_j and u_j , whereas subobserver i is influenced by its neighboring subobserver j through $\hat{x}_j$. Thus, there exists internal interaction among subsystems/subobservers.
- 2) *Bidirectional interaction*, which exists *between* the interconnected system and its associated interconnected observer. From (6) and (8), it can be seen that the optimal FE performance is affected by the interconnected system via the interconnection $D_{ij} x_j$. Meanwhile, the optimal FTC performance is affected by the interconnected observer via the estimation error e_i and the interconnection $B_i \bar{K}_{ij} \hat{x}_j$.

Thus, there exists bidirectional interaction between these two sides. The aforementioned interactions, indeed, break down the classical separation principle; however, they also provide a new clue to design a distributed optimal FE and FTC scheme.

Problem: Consider the interconnected system (1) with fault (2); the objectives of this article are twofold, which are as follows.

- Establish a framework to investigate the relationship between the interconnected system and its associated diagnostic interconnected observer under the internal and bidirectional interaction.

- Under the proposed framework, design a distributed optimal FE and further make use of it to design the distributed optimal fault-tolerant controller, such that the state z_i of system (1) asymptotically converges to zero. Furthermore, both the distributed optimal FE and FTC performances can be achieved.

III. STACKELBERG DIFFERENTIAL GRAPHICAL GAME-BASED DISTRIBUTED OPTIMAL FE AND FTC

In this section, a framework of the Stackelberg differential graphical game is established for the interconnected observer and interconnected system to achieve the distributed optimal FE and FTC.

A. Stackelberg Differential Graphical Game

In the Stackelberg differential graphical game, the leader and the follower are composed of several players, which form two graphical games. Suppose the strategies of players in the leader and the follower are $\mu \triangleq \{\mu_1, \mu_2, \dots, \mu_m\}$, $\forall \mu_i \in \mathcal{U}_i$, and $v \triangleq \{v_1, v_2, \dots, v_m\}$, $\forall v_i \in \mathcal{V}_i$, where $\mathcal{U}_i$ and $\mathcal{V}_i$ are admissible sets of strategies μ_i and v_i , respectively. The cost functions for the leader and the follower are assumed to be $\bar{\Gamma} \triangleq \{\bar{\Gamma}_1, \bar{\Gamma}_2, \dots, \bar{\Gamma}_m\}$ and $\bar{J} \triangleq \{\bar{J}_1, \bar{J}_2, \dots, \bar{J}_m\}$, respectively. The sets of neighboring strategies are represented as $\mu_{-i} \triangleq \{\mu_j : j \in N_i\}$ and $v_{-i} \triangleq \{v_j : j \in N_i\}$. The definitions of the best response are given as follows.

Definition 1 (Best response): In the Stackelberg differential graphical game, the best response of the follower is that for fixed μ_i , μ_{-i} , and v_{-i} , there exists v_i^* such that

$$\bar{J}_i(\mu_i, \mu_{-i}, v_i^*(\mu_i), v_{-i}(\mu_{-i})) \leq \bar{J}_i(\mu_i, \mu_{-i}, v_i(\mu_i), v_{-i}(\mu_{-i})). \quad (9)$$

The best response of the leader is that for fixed μ_{-i} , v_i and v_{-i} , there exists μ_i^* such that

$$\bar{\Gamma}_i(\mu_i^*, \mu_{-i}, v_i(\mu_i), v_{-i}(\mu_{-i})) \leq \bar{\Gamma}_i(\mu_i, \mu_{-i}, v_i(\mu_i), v_{-i}(\mu_{-i})). \quad (10)$$

□

This definition implies that the best response is the optimal strategy of a player, which takes into account other players' strategies to minimize its own cost. Inspired from the work in [15], the equilibrium for Stackelberg differential graphical game is defined.

Definition 2 (Interactive Stackelberg Equilibrium): The pair $\{\mu^*, v^*(\mu^*)\}$ is an interactive Stackelberg equilibrium with respect to $\bar{J}_i$ and $\bar{\Gamma}_i$, $\forall i \in \mathcal{M}$, if the following properties are satisfied.

- 1) For each $\mu_i \in \mathcal{U}_i$ and $\mu_{\mathcal{G}-i} \in \mathcal{U}_{\mathcal{G}-i}$, there exists $v_i^*(\mu_i) \in \mathcal{V}_i$ for the follower such that

$$\bar{J}_i(\mu_i, \mu_{\mathcal{G}-i}, v_i^*(\mu_i), v_{\mathcal{G}-i}^*(\mu_{\mathcal{G}-i})) \leq \bar{J}_i(\mu_i, \mu_{\mathcal{G}-i}, v_i(\mu_i), v_{\mathcal{G}-i}^*(\mu_{\mathcal{G}-i})) \quad (11)$$

where $\mathcal{G} - i \triangleq \{j : j \in \mathcal{M}, j \neq i\}$ is a set of other nodes in the graph $\mathcal{G}$ except node i .

- 2) There exists $\mu_i^* \in \mathcal{U}_i$ on the best responses of the leader such that

$$\bar{\Gamma}_i(\mu_i^*, \mu_{\mathcal{G}-i}^*, v_i^*(\mu_i^*), v_{\mathcal{G}-i}^*(\mu_{\mathcal{G}-i}^*)) \leq \bar{\Gamma}_i(\mu_i, \mu_{\mathcal{G}-i}^*, v_i^*(\mu_i), v_{\mathcal{G}-i}^*(\mu_{\mathcal{G}-i}^*)) \quad (12)$$

for any pair $(\mu, v(\mu))$ on the best responses of the leader. □

Different from the standard definition of Stackelberg equilibrium [16], there are two Nash equilibriums in the leader and follower shown in (11) and (12), respectively. Combining these two Nash equilibriums constitutes an interactive Stackelberg equilibrium, which can be regarded as an extension of the standard one. Based on the aforementioned definitions, next we show that the distributed optimal FE and FTC problem can be transformed into a Stackelberg differential graphical game.

B. Game Setting for Distributed Optimal FE and FTC

To construct a complete game situation, a new subobserver is designed to replace the subobserver in (4), in which an auxiliary controller

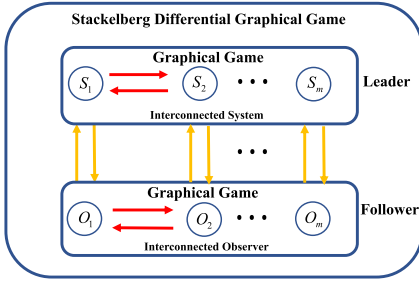


Fig. 1. Stackelberg differential graphical game for the interconnected system and interconnected observer (the red arrow represents the internal interaction; the yellow arrow represents the bidirectional interaction).

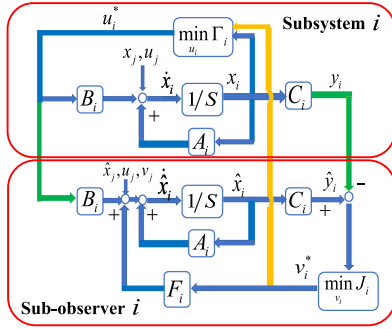


Fig. 2. Sequential decision of the interconnected system and interconnected observer.

$v_i \in R^{q_i}$ is designed for the subobserver i as follows:

$$O_i : \begin{cases} \dot{\hat{x}}_i = A_i \hat{x}_i + B_i u_i + \sum_{j \in N_i} B_{ij} u_j + F_i v_i + \sum_{j \in N_i} F_{ij} v_j \\ \quad + \sum_{j \in N_i} D_{ij} \hat{x}_j \\ \hat{y}_i = C_i \hat{x}_i \end{cases} \quad (13)$$

where $v_j \in R^{q_j}$ is the auxiliary controller of the neighboring subobserver j , $F_i \in R^{(n_i+n_{f_i}) \times q_i}$ and $F_{ij} \in R^{(n_i+n_{f_i}) \times q_j}$ are the input and coupling matrices, respectively. Due to the internal interaction, the interconnected system/observer can be formulated as two graphical games: Each subsystem/subobserver is viewed as a player in the graphical game, and the interconnections among subsystems/subobservers are regarded as edges of the graphical game such that all edges contribute to a connected graph $\mathcal{G}$, whose diagram is presented in Fig. 1. Next, we will show that the distributed optimal FE problem of the interconnected observer (13) and the distributed optimal FTC problem of the interconnected system (3) can be formulated as a Stackelberg differential graphical game because of the following “sequential” characteristics.

- *The sequence of existence:* The interconnected observer aims to observe the states/faults of the interconnected system. Thus, the existence of the interconnected observer depends on the corresponding interconnected system.
- *The sequence of strategy making:* Taken subsystem i and its associated subobserver i as an example in Fig. 2, it is shown that the subsystem operates an initial strategy u_i (may not optimal at the beginning), which can be used by subobserver i . After the subobserver receives the information of u_i and y_i from the subsystem (green line), the subobserver will make the best response to the subsystem’s current strategy, i.e., designing the optimal auxiliary controller v_i^* to minimize $e_{yi} \triangleq y_i - \hat{y}_i$. In return, the subsystem adjusts its own strategy based on the state and FE from the subobserver (yellow line), i.e., designing the optimal fault-tolerant controller u_i^* .

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One can see from the previous features that in the decision process, the subsystem has the dominant position, which not only forces its decision on the subobserver but also adjusts itself according to the subobserver’s strategy. Thus, the roles of leader and follower are assigned to the interconnected system (3) and the interconnected observer (13), respectively, as shown in Fig. 1.

It is worth mentioning that the design of the auxiliary controller v_i is the key to constructing the complete game situation in the Stackelberg differential graphical game: On the one hand, this auxiliary controller v_i cooperates with its neighbors’ auxiliary controllers v_j to achieve the Nash equilibrium in the interconnected observer. On the other hand, it simultaneously noncooperates with controllers $u_i, \forall i \in \mathcal{M}$ of the interconnected system to achieve the interactive Stackelberg equilibrium.

In the following, the distributed optimal FE and FTC designs for the interconnected observer (follower) and the interconnected system (leader) are presented in Sections IV and V, respectively.

IV. FOLLOWER’S DESIGN

In this section, the interconnected observer is regarded as the follower in the Stackelberg differential graphical game to achieve the distributed optimal FE under an arbitrary strategy of the leader, $u_i \in U_i, \forall i \in \mathcal{M}$, where U_i is the admissible set of controllers.

A. Distributed Optimal FE Problem for the Follower

The cost function for subobserver i in (13) is defined as

$$J_i(v_i, v_{-i}) \triangleq \int_0^\infty \left\{ \left(\hat{Y}_i - Y_i \right)^T Q_i \left(\hat{Y}_i - Y_i \right) + v_i^T R_i v_i + \sum_{j \in N_i} v_j^T R_{ij} v_j + u_i^T G_i v_i + \sum_{j \in N_i} u_j^T G_{ij} v_j \right\} dt \quad (14)$$

where $Y_i \triangleq [y_i^T \ y_{-i}^T]^T$ is the aggregated output of both subobserver i and its neighbors, $\hat{Y}_i$ is the estimate of Y_i . The matrices $Q_i \triangleq \begin{bmatrix} Q_{ii}^{(i)} & Q_{i,-i}^{(i)} \\ (Q_{i,-i}^{(i)})^T & Q_{-i,-i}^{(i)} \end{bmatrix}$, R_i , R_{ij} , G_i , and G_{ij} are symmetric positive definite. The term $u_i^T G_i v_i$ represents the impact from subsystem i to subobserver i , which results from the bidirectional interaction. The term $u_j^T G_{ij} v_j$ represents the coupling effect from the neighbors.

According to the analysis in Section III-B, subsystem i forces its controller u_i on subobserver i . Thus, we first investigate the optimization problem of the follower under the arbitrary strategies of the leader $u_i \in U_i, u_{-i} \in U_{-i}$, which is described as

$$\min_{v_i} J_i(v_i, v_{-i}) \quad \text{for given } u_i, u_{-i}. \quad \text{s.t.} \quad (13)$$

In the following, it will be shown that if the optimal solution v_i^* can be obtained, then the estimation error e_{xi} converges to zero asymptotically.

Remark 4: Note that the leader’s controllers u_i and u_j are first assumed to be given and admissible when we investigate the follower’s problem in the framework of Stackelberg game [17]–[19]; thus, the follower’s optimization problem with the cost function (14) can be regarded as a standard linear quadratic tracking control problem by treating u_i and u_j as exogenous signals. The terms $u_i^T G_i v_i$ and $u_j^T G_{ij} v_j$ may be negative during the dynamical process when the strategies u_i (u_j) and v_i (v_j) have opposite signs, which is a result of the noncooperation between the interconnected system and the interconnected observer. The cost function J_i in (14) is appropriate in the sense that the cost function J_i is continuous and convex with respect to v_i and v_j , whose proof is presented in [19]. $\square$

B. Solution of the Follower

To solve the follower’s optimization problem, the Hamiltonian function for subobserver i is defined as

$$H_{io}(\hat{X}_i, Y_i, \nabla J_i, v_i, v_{-i})$$

$$\begin{aligned}
& \triangleq (\tilde{C}_i \hat{X}_i - Y_i)^T Q_i (\tilde{C}_i \hat{X}_i - Y_i) + v_i^T R_i v_i + \sum_{j \in N_i} v_j^T R_{ij} v_j \\
& + u_i^T G_i v_i + \sum_{j \in N_i} u_j^T G_{ij} v_j + \nabla J_i^T (A_i \hat{x}_i + B_i u_i \\
& + \sum_{j \in N_i} B_{ij} u_j + F_i v_i + \sum_{j \in N_i} F_{ij} v_j + \sum_{j \in N_i} D_{ij} \hat{x}_j) \quad (15)
\end{aligned}$$

where $\tilde{C}_i \triangleq \text{diag}(C_i, C_{-i})$, $\hat{X}_i \triangleq [\hat{x}_i^T \hat{x}_{-i}^T]^T$, $\hat{x}_{-i} \triangleq \{\hat{x}_j : j \in N_i\}$ is the aggregated state vector of the neighboring subobservers, and $\nabla J_i \triangleq \partial J_i / \partial \hat{x}_i$ means the partial derivative of the cost function J_i with respect to the estimate $\hat{x}_i$.

The optimal value J_i^* of the cost function (14) satisfies the coupled Hamilton–Jacobi (HJ) equation

$$0 = \min_{v_i} H_{i0}(\hat{X}_i, Y_i, \nabla J_i^*, v_i, v_{-i}^*) \quad \forall i \in \mathcal{M}. \quad (16)$$

Based on the minimal principle [15], [30], the necessary condition for solving (16) is $\nabla \dot{J}_i = -\frac{\partial H_{i0}}{\partial \hat{x}_i}$ such that

$$\begin{aligned}
\nabla \dot{J}_i = & -A_i^T \nabla J_i - 2C_i^T Q_{ii}^{(i)} C_i \hat{x}_i + 2C_i^T Q_{ii}^{(i)} y_i \\
& - 2C_i^T Q_{i,-i}^{(i)} C_{-i} \hat{x}_{-i} + 2C_i^T Q_{i,-i}^{(i)} y_{-i}. \quad (17)
\end{aligned}$$

The extreme condition is $\frac{\partial H_{i0}}{\partial v_i} = 0$ such that $2R_i v_i + G_i u_i + F_i^T \nabla J_i = 0$, which implies that the optimal auxiliary controller is

$$v_i^* = -\frac{1}{2} R_i^{-1} (F_i^T \nabla J_i^* + G_i u_i). \quad (18)$$

Substituting (18) into (16) leads to the follower's coupled HJ equations

$$\begin{aligned}
& (\tilde{C}_i \hat{X}_i - Y_i)^T Q_i (\tilde{C}_i \hat{X}_i - Y_i) + (v_i^*)^T R_i v_i^* + \sum_{j \in N_i} (v_j^*)^T R_{ij} v_j^* \\
& + u_i^T G_i v_i^* + \sum_{j \in N_i} u_j^T G_{ij} v_j^* + (\nabla J_i^*)^T (A_i \hat{x}_i + B_i u_i \\
& + \sum_{j \in N_i} B_{ij} u_j + F_i v_i^* + \sum_{j \in N_i} F_{ij} v_j^* + \sum_{j \in N_i} D_{ij} \hat{x}_j) = 0 \quad (19)
\end{aligned}$$

with $J_i^*(0) = 0$, $\forall i \in \mathcal{M}$. Based on the designed optimal auxiliary controller in (18), the convergence characteristic of the estimation error, and the equilibrium of the whole interconnected observer are analyzed in the following theorem.

Theorem 1: Suppose Assumption 1 holds. Given $u_i \in U_i$, $u_{-i} \in U_{-i}$, the optimal auxiliary controller v_i^* designed in (18) with the solutions ∇J_i^* from the HJ equations (19). It holds the following.

- 1) $\lim_{t \rightarrow \infty} (x_i - \hat{x}_i) = 0$.
- 2) The m-tuple of auxiliary controllers $(v_1^*, v_2^*, \dots, v_m^*)$ constitutes the Nash equilibrium for the graphical game in the interconnected observer, and the corresponding game value is $J_i(\hat{X}_i(0), Y_i(0))$, $i \in \mathcal{M}$. $\square$

Proof: 1) Choose J_i^* as the Lyapunov function. Making use of the HJ equations in (19), the time derivative of J_i^* can be obtained

$$\begin{aligned}
\dot{J}_i^* &= \left(\frac{\partial J_i^*}{\partial \hat{x}_i} \right)^T \dot{\hat{x}}_i \\
&= -(\hat{Y}_i - Y_i)^T Q_i (\hat{Y}_i - Y_i) - (v_i^*)^T R_i v_i^* \\
&\quad - \sum_{j \in N_i} (v_j^*)^T R_{ij} v_j^* - u_i^T G_i v_i^* - \sum_{j \in N_i} u_j^T G_{ij} v_j^*. \quad (20)
\end{aligned}$$

Note that the leader's controllers u_i and u_j are exogenous signals with respect to the follower; we proceed to prove that $(\hat{Y}_i - Y_i)$ converges to zero if u_i and u_j converge to zero. By substituting the optimal auxiliary controller v_i^* in (18) into $-(v_i^*)^T R_i v_i^* - u_i^T G_i v_i^*$ and making some algebraic manipulations, we obtain

$$\begin{aligned}
& -(v_i^*)^T R_i v_i^* - u_i^T G_i v_i^* \\
&= -\frac{1}{4} (\nabla J_i^*)^T F_i R_i^{-1} F_i^T \nabla J_i^* + \frac{1}{4} u_i^T G_i R_i^{-1} G_i u_i. \quad (21)
\end{aligned}$$

Similarly, we have

$$\begin{aligned}
& -(v_j^*)^T R_{ij} v_j^* - u_j^T G_{ij} v_j^* \\
&= -\frac{1}{4} (\nabla J_j^*)^T F_j R_j^{-1} R_{ij} R_j^{-1} F_j^T \nabla J_j^* + \frac{1}{2} (\nabla J_j^*)^T F_j R_j^{-1} G_{ij} u_j \\
&\quad - \frac{1}{2} (\nabla J_j^*)^T F_j R_j^{-1} R_{ij} R_j^{-1} G_j u_j - \frac{1}{4} u_j^T G_j R_j^{-1} R_{ij} R_j^{-1} G_j u_j \\
&\quad + \frac{1}{2} u_j^T G_{ij} R_j^{-1} G_j u_j. \quad (22)
\end{aligned}$$

By letting $R_{ij} R_j^{-1} G_j = G_{ij}$, (22) can be written as

$$\begin{aligned}
& -(v_j^*)^T R_{ij} v_j^* - u_j^T G_{ij} v_j^* \\
&= -\frac{1}{4} (\nabla J_j^*)^T F_j R_j^{-1} R_{ij} R_j^{-1} F_j^T \nabla J_j^* + \frac{1}{4} u_j^T G_{ij} R_j^{-1} G_j u_j. \quad (23)
\end{aligned}$$

Then, substituting (21) and (23) into (20) leads to

$$\begin{aligned}
\dot{J}_i^* &= -(\hat{Y}_i - Y_i)^T Q_i (\hat{Y}_i - Y_i) - \frac{1}{4} (\nabla J_i^*)^T F_i R_i^{-1} F_i^T \nabla J_i^* \\
&\quad - \frac{1}{4} (\nabla J_j^*)^T F_j R_j^{-1} R_{ij} R_j^{-1} F_j^T \nabla J_j^* + \frac{1}{4} u_i^T G_i R_i^{-1} G_i u_i \\
&\quad + \frac{1}{2} u_j^T G_{ij} R_j^{-1} G_j u_j \quad (24)
\end{aligned}$$

which indicates that for any $(\hat{Y}_i - Y_i)^T Q_i (\hat{Y}_i - Y_i) \geq \frac{1}{4} u_i^T G_i R_i^{-1} G_i u_i + \frac{1}{4} u_j^T G_{ij} R_j^{-1} G_j u_j$, the following inequality

$$\begin{aligned}
\dot{J}_i^* &\leq -\frac{1}{4} (\nabla J_i^*)^T F_i R_i^{-1} F_i^T \nabla J_i^* - \frac{1}{4} (\nabla J_j^*)^T F_j R_j^{-1} R_{ij} R_j^{-1} F_j^T \nabla J_j^* \\
&\leq -\frac{1}{4} (\nabla J_i^*)^T F_i R_i^{-1} F_i^T \nabla J_i^* \quad (25)
\end{aligned}$$

is satisfied. Based on [27, Th. C.2, p.502], we have $\|\hat{Y}_i(t) - Y_i(t)\| \leq \eta_i(\|\hat{Y}_i(0) - Y_i(0)\|, t) + \delta_i(\|u_i(t)\|, \|u_j(t)\|)$, $\forall t \geq 0$, where η_i is a class of $\mathcal{KL}$ function and δ_i is a class of $\mathcal{K}$ function. Note that u_i and u_j are admissible controllers to stabilize the system of the leader [15], implying that $u_i \rightarrow 0$ and $u_j \rightarrow 0$ as $t \rightarrow \infty$. Thus, it follows that $(\hat{Y}_i - Y_i) \rightarrow 0$ as $t \rightarrow \infty$. Furthermore, under the observability conditions in Assumption 1, it is proved in [29] that $\hat{X}_i$ approximates to X_i asymptotically. That holds for all $i \in \mathcal{M}$, such that the estimate error $\hat{x}_i - x_i$ converges to zero asymptotically.

2) The cost function J_i in (14) can be expressed as

$$\begin{aligned}
J_i(\hat{X}_i(0), Y_i(0), v_i, v_{-i}) &= \int_0^\infty \left((\tilde{C}_i \hat{X}_i - Y_i)^T Q_i (\tilde{C}_i \hat{X}_i - Y_i) + v_i^T R_i v_i + \sum_{j \in N_i} v_j^T R_{ij} v_j \right. \\
&\quad \left. + u_i^T G_i v_i + \sum_{j \in N_i} u_j^T G_{ij} v_j \right) dt + \int_0^\infty (\nabla J_i)^T (A_i \hat{x}_i + B_i u_i \\
&\quad + F_i v_i + \sum_{j \in N_i} B_{ij} u_j + \sum_{j \in N_i} F_{ij} v_j + \sum_{j \in N_i} D_{ij} \hat{x}_j) dt \\
&\quad + J_i(\hat{X}_i(\infty), Y_i(\infty)) - J_i(\hat{X}_i(0), Y_i(0)). \quad (26)
\end{aligned}$$

According to the proof of part 1), J_i converges to zero as $t \rightarrow \infty$, i.e., $J_i(\hat{X}_i(\infty), Y_i(\infty)) = 0$. Then, by using the HJ equation in (19) and completing the squares of (26) lead to

$$\begin{aligned}
J_i(\hat{X}_i(0), Y_i(0), v_i, v_{-i}) &= \int_0^\infty \left((v_i - v_i^*)^T R_i (v_i - v_i^*) + \sum_{j \in N_i} (v_j - v_j^*)^T R_{ij} (v_j - v_j^*) \right. \\
&\quad \left. + 2 \sum_{j \in N_i} (v_j^*)^T R_{ij} (v_j - v_j^*) + \nabla J_i^T \sum_{j \in N_i} F_{ij} (v_j - v_j^*) \right. \\
&\quad \left. + \sum_{j \in N_i} u_j^T G_{ij} (v_j - v_j^*) \right) dt + J_i(\hat{X}_i(0), Y_i(0)). \quad (27)
\end{aligned}$$

Suppose that the neighboring subobservers are optimal, i.e., $v_j = v_j^*$, $j \in N_i$, the cost function in (27) becomes

$$J_i(v_i, v_{-i}^*) = \int_0^\infty (v_i - v_i^*)^T R_i (v_i - v_i^*) dt + J_i(\hat{X}_i(0), Y_i(0)). \quad (28)$$

Setting $v_i = v_i^*$ leads to the optimal value, i.e.,

$$J_i^*(v_i^*, v_{-i}^*) = J_i(\hat{X}_i(0), Y_i(0)). \quad (29)$$

Since (29) $\leq$ (28), we obtain that for given u_i and u_{-i}

$$J_i(u_i, u_{-i}, v_i^*, v_{-i}^*) \leq J_i(u_i, u_{-i}, v_i, v_{-i}^*). \quad (30)$$

Thus, the condition in (9) is satisfied. Moreover, the condition (30) is satisfied for all $i \in \mathcal{M}$, such that the condition of Nash equilibrium in (11) is achieved. ■

Theorem 1 achieves the distributed optimal FE goal of the interconnected observer. Due to the internal interaction, all subobservers cooperate to minimize their own cost functions as well as improving their neighbors' performance, which helps to achieve Nash equilibrium in the graphical game. Meanwhile, the subobserver's policy v_i^* is influenced by the leader's strategy u_i shown in (18), which results from the bidirectional interaction. Based on the distributed optimal FE, the distributed optimal FTC problem for the interconnected system will be solved next.

V. LEADER'S DESIGN

In this section, the interconnected system is regarded as the leader in the Stackelberg differential graphical game to achieve the distributed optimal FTC.

A. Distributed Optimal FTC Problem for the Leader

The cost function for subsystem i is defined as

$$\Gamma_i(u_i, u_{-i}, v_i) \triangleq \int_0^\infty (X_i^T M_i X_i + u_i^T L_i u_i + \sum_{j \in N_i} u_j^T L_{ij} u_j + v_i^T \Psi_i v_i) dt \quad (31)$$

where $X_i \triangleq [x_i^T x_{-i}^T]^T$, $x_{-i} \triangleq \{x_j : j \in N_i\}$ is the aggregated state vector of the neighboring subsystems, the matrices $M_i \triangleq \begin{bmatrix} M_{ii}^{(i)} & M_{i,-i}^{(i)} \\ (M_{i,-i}^{(i)})^T & M_{-i,-i}^{(i)} \end{bmatrix}$, L_i , L_{ij} , and Ψ_i are symmetric positive definite.

Under the cost function (31), the optimization problem for the leader is

$$\min_{u_i} \Gamma_i(u_i, u_{-i}, v_i^*) \quad \text{s.t. (3) and (17).}$$

Two constraints are considered in the leader's optimization problem: One is the dynamics of subsystem i in (3); another is the dynamics of ∇J_i in (17), which is the unique influential factor of v_i . Note that the best response strategy v_i^* of the follower is used for solving the leader's optimization problem in the framework of a Stackelberg differential game [17]–[19]. In this way, the leader's action takes into account the current best response v_i^* of the follower such that the FTC performance of the subsystem can be adjusted based on the FE performance of the subobserver, which further reveals the leader's role of the interconnected system.

B. Solution of the Leader

According to the minimal principle, the corresponding Hamiltonian function for subsystem i is defined as

$$\begin{aligned} H_{is}(X_i, \nabla \Gamma_i, \beta_i, \nabla J_i, u_i, u_{-i}, v_i^*) \\ \triangleq X_i^T M_i X_i + u_i^T L_i u_i + \sum_{j \in N_i} u_j^T L_{ij} u_j + (v_i^*)^T \Psi_i v_i^* + \nabla \Gamma_i^T (A_i x_i \\ + B_i u_i + \sum_{j \in N_i} B_{ij} u_j + \sum_{j \in N_i} D_{ij} x_j) + \beta_i^T (-A_i^T \nabla J_i - 2C_i^T Q_{ii}^{(i)} C_i \hat{x}_i \\ + 2C_i^T Q_{ii}^{(i)} y_i - 2C_i^T Q_{i,-i}^{(i)} C_{-i} \hat{x}_{-i} + 2C_i^T Q_{i,-i}^{(i)} y_{-i}) \end{aligned}$$

where $\nabla \Gamma_i \triangleq \partial \Gamma_i / \partial x_i$ and $\beta_i \triangleq \partial \Gamma_i / \partial (\nabla J_i)$. The optimal value Γ_i^* of the cost function (31) satisfies the coupled HJ equation

$$0 = \min_{u_i} H_{is}(X_i, \nabla \Gamma_i^*, \beta_i^*, \nabla J_i^*, u_i, u_{-i}, v_i^*) \quad \forall i \in \mathcal{M}. \quad (32)$$

To solve (32), the necessary conditions are given in $\nabla \dot{\Gamma}_i = -\frac{\partial H_{is}}{\partial x_i}$ and $\dot{\beta}_i = -\frac{\partial H_{is}}{\partial (\nabla J_i)}$, which lead to

$$\nabla \dot{\Gamma}_i = -2M_{ii}^{(i)} x_i - 2M_{i,-i}^{(i)} x_{-i} - A_i^T \nabla \Gamma_i - 2C_i^T Q_{ii}^{(i)} C_i \beta_i \quad (33)$$

$$\dot{\beta}_i = A_i \beta_i - \left(\frac{1}{2} \Delta_i - \frac{1}{2} W_i \Pi_i^{-1} W_i^T \right) \nabla J_i^* + \frac{1}{2} W_i \Pi_i^{-1} B_i^T \nabla \Gamma_i \quad (34)$$

where $W_i \triangleq F_i R_i^{-1} \Psi_i R_i^{-1} G_i$ and $\Delta_i \triangleq F_i R_i^{-1} \Psi_i R_i^{-1} F_i^T$. The extreme condition of $0 = \frac{\partial H_{is}}{\partial u_i}$ leads to $B_i^T \nabla \Gamma_i + 2L_i u_i + \frac{1}{2} G_i R_i^{-1} \Psi_i R_i^{-1} G_i u_i + \frac{1}{2} G_i R_i^{-1} \Psi_i R_i^{-1} F_i^T \nabla J_i^* = 0$, from which the distributed optimal fault-tolerant controller can be obtained

$$u_i^* = -\Pi_i^{-1} B_i^T \nabla \Gamma_i^* - \frac{1}{2} \Pi_i^{-1} W_i^T \nabla J_i^* \quad (35)$$

where $\Pi_i \triangleq 2L_i + \frac{1}{2} G_i R_i^{-1} \Psi_i R_i^{-1} G_i$. Two deep insights are provided in the following to reveal the properties of u_i^* .

- (i) The reconfiguration of the fault-tolerant controller in (35) is based on the state and FE $\hat{X}_i$, which coincides with the core idea of active FTC. From (33) and (34), we find that $\nabla \Gamma_i$ is the function of X_i and β_i , whereas β_i is the function of ∇J_i and $\nabla \Gamma_i$. At the optimal point, ∇J_i^* has the form of $\nabla J_i^* = P_i \hat{X}_i + g_i$, where P_i is a feedback matrix in the Riccati equation, and g_i is a function related to the output Y_i [30]. In summary, $\nabla \Gamma_i^* = \varphi_{i1}(X_i, \beta_i) \xrightarrow{\nabla J_i^* = P_i \hat{X}_i + g_i} \xrightarrow{\beta_i = \varphi_{i2}(\nabla J_i^*, \nabla \Gamma_i^*)} \nabla \Gamma_i^* = \varphi_{i3}(X_i, \hat{X}_i, Y_i)$. Therefore, the estimate $\hat{X}_i$ is used to construct the optimal fault-tolerant controller u_i^* . However, since the cost function penalizes the state X_i and forces the real state to converge to zero, the real state X_i appears inevitably in the controller u_i^* . Thus, the estimated state $\hat{X}_i$ should be used to replace X_i when we solve the leader's coupled HJ equation (which will be derived later).
- (ii) The optimal fault-tolerant controller u_i^* is distributed. From (33), it can be found that $\nabla \Gamma_i$ is a function of x_i and x_{-i} , such that $\nabla \Gamma_i^*$ can be expressed as $\nabla \Gamma_i^*(X_i) = \nabla \Gamma_{ii}^*(x_i) + \sum_{j \in N_i} \nabla \Gamma_{ij}^*(x_i, x_j)$, where Γ_{ii}^* is the cost for the subsystem i 's self-dynamics, and Γ_{ij}^* is the cost for compensating the interconnection [14]. Therefore, u_i^* can be further expressed in a distributed manner as $u_i^* = -\Pi_i^{-1} B_i^T \nabla \Gamma_{ii}^* - \Pi_i^{-1} B_i^T \sum_{j \in N_i} \nabla \Gamma_{ij}^* - \frac{1}{2} \Pi_i^{-1} W_i^T \nabla J_i^*$. □

Then, the coupled HJ equations for the leader can be derived as

$$\begin{aligned} X_i^T M_i X_i + (u_i^*)^T L_i u_i^* + \sum_{j \in N_i} (u_j^*)^T L_{ij} u_j^* + (v_i^*)^T \Psi_i v_i^* + (\nabla \Gamma_i^*)^T \\ (A_i x_i + B_i u_i^* + \sum_{j \in N_i} B_{ij} u_j^* + \sum_{j \in N_i} D_{ij} x_j) + \beta_i^T (-A_i^T \nabla J_i^* \\ - 2C_i^T Q_{ii}^{(i)} C_i \hat{x}_i + 2C_i^T Q_{ii}^{(i)} y_i - 2C_i^T Q_{i,-i}^{(i)} C_{-i} \hat{x}_{-i} \\ + 2C_i^T Q_{i,-i}^{(i)} y_{-i}) = 0 \end{aligned} \quad (36)$$

with $\Gamma_i^*(0) = 0, \forall i \in \mathcal{M}$.

Under the optimal FTC controller u_i^* in (35), we are going to analyze the stability and Nash equilibrium of the faulty interconnected system (3).

Theorem 2: Suppose Assumption 1 holds. The distributed optimal fault-tolerant controllers u_i^* is designed in (35) with solutions $\nabla \Gamma_i^*$ and ∇J_i^* from the coupled HJ equations (19) and (36). Then, the following statements hold.

- 1) The faulty interconnected system (3) is asymptotically stable.

- 2) The m-tuple of controllers $(u_1^*, u_2^*, \dots, u_m^*)$ constitutes Nash equilibrium for the graphical game in the interconnected system, and the corresponding game value is $\Gamma_i^*(X_i(0), \nabla J_i(0))$, $i \in \mathcal{M}$. $\square$

Proof: 1) Choose Γ_i^* as the Lyapunov function. Making use of the coupled HJ equation in (36), the time derivative of Γ_i^* is

$$\begin{aligned} \dot{\Gamma}_i^* &= \left(\frac{\partial \Gamma_i^*}{\partial x_i} \right)^T \dot{x}_i + \left(\frac{\partial \Gamma_i^*}{\partial (\nabla J_i)} \right)^T \nabla \dot{J}_i \\ &= -X_i^T M_i X_i - (u_i^*)^T L_i u_i^* - (u_j^*)^T L_{ij} u_j^* - (v_i^*)^T \Psi_i v_i^*. \end{aligned}$$

Due to $M_i > 0$, $L_i > 0$, $L_{ij} > 0$, and $\Psi_i > 0$, one has $\dot{\Gamma}_i^* \leq 0$, which means that the faulty interconnected system (3) is asymptotically stable by using u_i^* .

2) This proof is similar to the proof of part 2) in Theorem 1, and is, therefore, omitted. $\blacksquare$

Theorem 2 achieves the distributed optimal FTC goal of the interconnected system, in which each subsystem has the initiative to adjust its controller u_i^* based on the follower's best response v_i^* . In this way, the FTC performance can be regulated according to the FE performance.

C. Interactive Stackelberg Equilibrium

To this end, we shall comprehensively analyze the FE and FTC performances. Under the distributed optimal fault-tolerant controller u_i^* , the follower's optimal strategy can be further expressed as

$$v_i^* = -\frac{1}{2} R_i^{-1} (F_i^T \nabla J_i^* + G_i u_i^*). \quad (37)$$

The following theorem shows that the leader's strategies $u^* \triangleq \{u_1^*, u_2^*, \dots, u_m^*\}$ and the follower's strategies $v^* \triangleq \{v_1^*, v_2^*, \dots, v_m^*\}$ constitute an interactive Stackelberg equilibrium.

Theorem 3: Suppose Assumption 1 holds. The optimal control policies u_i^* and v_i^* are given by (35) and (37), respectively, $i \in \mathcal{M}$. Then, the pair $(u^*, v^*(u^*))$ reaches the interactive Stackelberg equilibrium of the Stackelberg differential graphical game between the interconnected system and its associated diagnostic interconnected observer. $\blacksquare$

Proof: According to Theorems 1 and 2, Nash equilibriums in the graphical games of leader and follower are both achieved, which implies that

$$\begin{aligned} &J_i(u_i, u_{G-i}, v_i^*(u_i), v_{G-i}^*(u_{G-i})) \\ &\leq J_i(u_i, u_{G-i}, v_i(u_i), v_{G-i}^*(u_{G-i})) \\ &\Gamma_i(u_i^*, u_{G-i}^*, v_i^*(u_i^*), v_{G-i}^*(u_{G-i}^*)) \\ &\leq \Gamma_i(u_i, u_{G-i}^*, v_i^*(u_i), v_{G-i}^*(u_{G-i}^*)). \end{aligned}$$

Therefore, two properties in Definition 2 are satisfied, which implies that the interactive Stackelberg equilibrium is achieved. $\blacksquare$

At the interactive Stackelberg equilibrium, the "competition" between the interconnected system and interconnected observer arrives at a balance point after several rounds of making the best response to each other, in which the interconnected observer achieves the optimal state and FE with minimal cost J_i^* , $\forall i \in \mathcal{M}$, whereas the interconnected system achieves the optimal FTC with the minimal cost Γ_i^* , $\forall i \in \mathcal{M}$. Furthermore, none of them would have intentions to leave this equilibrium, otherwise, both would suffer loss.

Based on Theorems 1–3, Algorithm 1 is presented to show the steps to carry out the Stackelberg differential graphical game based-active FTC scheme.

As shown in Theorem 3, the derivations of u_i^* and v_i^* depend on the solutions ∇J_i and $\nabla \Gamma_i$ in the coupled HJ equations (19) and (36). Since these coupled HJ equations are not only partial differential equations but also coupled with each other, it is usually difficult to solve them. ADP methodology can help us to approximate these solutions [14], [31]. It is worth mentioning that the estimated state $\hat{X}_i$ should be injected into the HJ equation in (36) through a hidden layer in order to replace the unknown state X_i , which helps to achieve the successful ADP-based Stackelberg differential graphical game strategies.

Remark 5: It has been proved in Theorem 1 that $\hat{X}_i$ converges to X_i asymptotically; thus, the controller u_i^* with the injection of state estimation $\hat{X}_i$ is still the optimal one for the leader, which guarantees the

Algorithm 1 Stackelberg differential graphical game-based-active FTC scheme

- Step 1: (Initialization)** Select appropriate matrices of Q_i , R_i , R_{ij} , G_i , G_{ij} , M_i , L_i , L_{ij} , and Ψ_i , $\forall i \in \mathcal{M}$, $j \in N_i$.
- Step 2: (Follower's design)** For each subobserver i , design the optimal auxiliary controller v_i^* in (18) by minimizing the cost function J_i under an arbitrary strategy of the leader, $u_i \in U_i$, $u_j \in U_j$, where U_i and U_j are the admissible sets of controllers.
- Step 3: (Leader's design)** For each subsystem i , design the optimal controller u_i^* in (35) by minimizing the cost function Γ_i under the constraints (3) and (17).
- Step 4: (Implementation)** Update v_i^* with u_i^* , and apply u_i^* in (35) and v_i^* in (37) into the subsystem i and the subobserver i , $\forall i \in \mathcal{M}$, respectively.
-

stability of the interconnected system, and satisfies the property in (12). Thus, according to Definition 2, the interactive Stackelberg equilibrium can still be achieved. A similar approach of using the estimation to replace the unknown state is used in [32], where the stability and H_∞ performance of the system are still remained after the replacement. It is worth pointing out that the replacement of the unavailable states and the approximation of the HJ equations can be integrated into the design of ADP technology [33], where the uniformly ultimately bounded of the system's state can be achieved, and an approximated equilibrium can be reached. $\square$

Remark 6: Apart from the FTC area, the Stackelberg differential graphical game approach can be utilized in other applications. For example, it can be used in air traffic control to deal with the leader–follower relationship between air-traffic management and pilots [34]. Another example is the control of unmanned air/ground vehicle (UAV–UGV) cooperative systems, in which UAVs and UGVs form two graphical games acting as the leader and the follower, respectively [35]. $\square$

VI. NUMERICAL EXAMPLE

In this section, an example of interconnected system is presented to illustrate the effectiveness of the proposed method. Consider the following faulty interconnected systems:

$$\bar{A}_1 = \begin{bmatrix} -0.5 & -1.5 \\ 2 & -3 \end{bmatrix}, \bar{A}_{f1} = \begin{bmatrix} -1 & 0.5 \\ 0 & -1.5 \end{bmatrix}, \bar{E}_1 \bar{C}_{f1} = \bar{E}_{21} \bar{C}_{f1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\bar{B}_{12} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \bar{D}_{12} = \begin{bmatrix} 1.5 & 1 \\ 0 & 2 \end{bmatrix}, \bar{C}_1 = \bar{C}_2 = [1 \ 0], \bar{A}_2 = \begin{bmatrix} -4.5 & 1.2 \\ 1 & -3.6 \end{bmatrix}$$

$$\bar{D}_{21} = \begin{bmatrix} 1.2 & 0.45 \\ 0 & 1 \end{bmatrix}, \bar{B}_{21} = \begin{bmatrix} 0.5 \\ 0 \end{bmatrix}, \bar{B}_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \bar{B}_{21} = \begin{bmatrix} 0.5 \\ 0 \end{bmatrix}, \bar{B}_1 = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$F_1 = [5 \ 4 \ 6 \ 4]^T, F_{12} = [2 \ 0 \ 1 \ 2]^T, F_2 = B_2, F_{21} = B_{21}.$$

The parameters in the cost functions (14) and (31) are chosen as

$$M_{11}^{(1)} = Q_{11}^{(1)} = I_{4 \times 4}, M_{12}^{(1)} = Q_{12}^{(1)} = 0.5 I_{4 \times 4}, M_{21}^{(2)} = Q_{21}^{(2)}$$

$$= 0.5 I_{4 \times 4}, M_{22}^{(2)} = Q_{22}^{(2)} = I_{4 \times 4}, \Psi_1 = \Psi_2 = I_{4 \times 4}, R_1 = 2,$$

$$R_2 = 3, R_{12} = L_{12} = 2R_{21} = G_2 = 3, G_1 = G_{12} = 2, G_{21} = 3,$$

$$L_1 = 2, L_2 = 5, L_{21} = 1.$$

Assume that the fault f_1 takes place at $t = 0.5$ s in subsystem 1, and the simulation results are displayed in Figs. 3 and 4. The state trajectories for subsystems 1 and 2 are shown in Fig. 3, which shows that the closed-loop interconnected system can maintain its stability after faults occur. Moreover, the estimation errors of the state and fault converge to zero, as shown in Fig. 4. As $x_i \rightarrow 0$, $e_i \rightarrow 0$, then $y_i \rightarrow 0$, $u_i \rightarrow 0$ and $v_i \rightarrow 0$. Thus, the optimal cost at the interactive Stackelberg equilibrium for the interconnected system and interconnected observer are $J_i^* = 0$ and $\Gamma_i^* = 0$. This verifies the effectiveness of the developed distributed optimal FE and FTC scheme for this example.

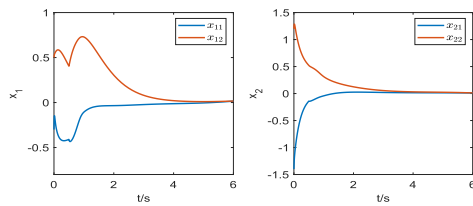


Fig. 3. State trajectories of subsystems 1 and 2.

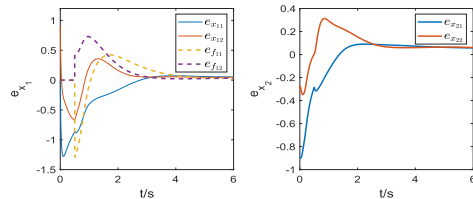


Fig. 4. Estimation errors of subsystems 1 and 2.

VII. CONCLUSION

This article develops a distributed optimal FE and FTC approach for the interconnected system based on the proposed Stackelberg differential graphical game, in which auxiliary controllers are designed for interconnected observer to complete the game situation. It is proved that the interconnected system and the interconnected observer reach the interactive Stackelberg equilibrium, at which the distributed optimal FE and FTC performances are both achieved. In our future work, more complex types of faults and malicious attacks will be considered, especially the case of unknown corrupted channels resulting from faults/attacks [36], which are crucial and urgent security issues for interconnected systems.

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